Drawing Structure List

ISO

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Revision History List

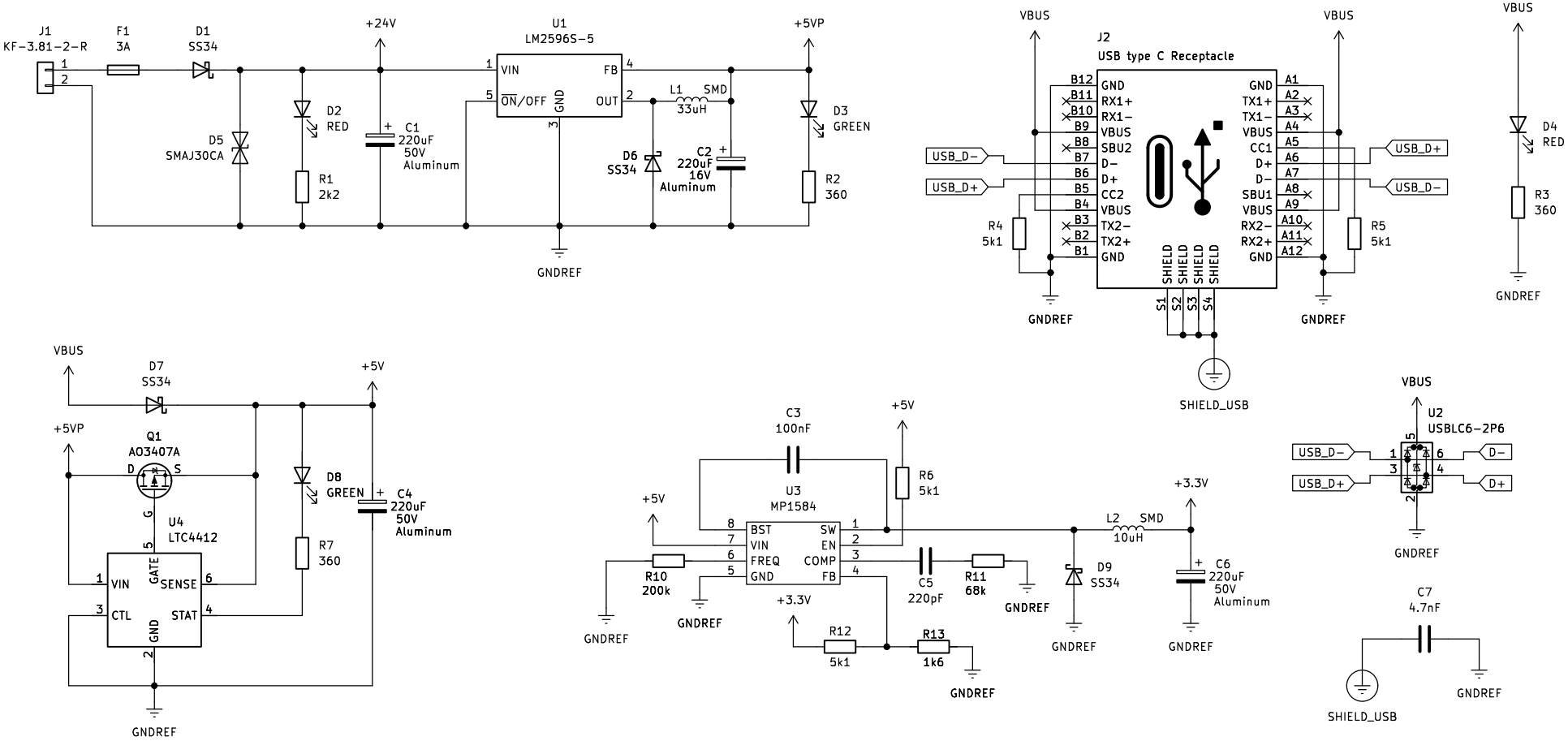
Revision	Page No.	Revision reasons and contents	PIC	Date
01	All	Make new	N. C. Binh	Mar.2026

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			N.C.BINH Mar. 2026	MODULE IO QI9-2604-A01	A	01

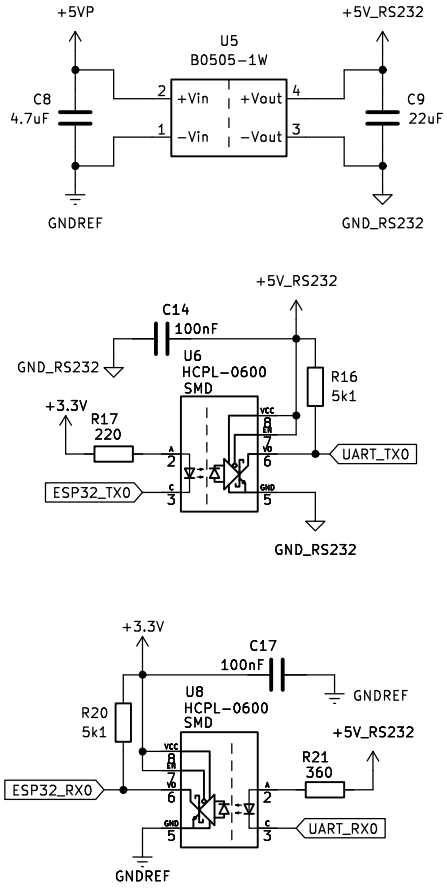
No.	Part Name	Part Code	Maker	Q'ty	Unit	Symbol	Remark
1	Polarized capacitor	220uF	Any	4	pcs	C1,C2,C4,C6	
2	Capacitor	100nF	Any	12	pcs	C3,C10,C14,C17,C18,C21,C22, C23,C24,C25,C26,C32	
3	Capacitor	220pF	Any	1	pcs	C5	
4	Capacitor	4.7nF	Any	3	pcs	C7,C11,C30	
5	Capacitor	4.7uF	Any	2	pcs	C8,C19	
6	Capacitor	22uF	Any	6	pcs	C9,C12,C13,C15,C16,C20	
7	Capacitor	1nF	Any	2	pcs	C27,C28	
8	Capacitor	22pF	Any	2	pcs	C29,C31	
9	Diode schottky	SS34	Any	4	pcs	D1,D6,D7,D9	
10	Led red	RED	Any	5	pcs	D2,D4,D11,D15,D19	
11	Led green	GREEN	Any	23	pcs	D3,D8,D20,D21,D22,D27,D28, D31,D32,D35,D36,D41,D42, D43,D44,D45,D46,D47,D50, D54,D56,D58,D61	
12	TVS diode	SMAJ30CA	LittelFuse	22	pcs	D5,D23,D24,D25,D26,D29,D30, D33,D34,D37,D38,D39,D40,D48, D49,D51,D52,D53,D55,D57,D59, D60	
13	Led yellow	YELLOW	Any	2	pcs	D10,D14	
14	TVS diode	SMAJ15CA	LittelFuse	2	pcs	D12,D13	
15	TVS diode	SMAJ9.0CA	LittelFuse	3	pcs	D16,D17,D18	
16	Fuse	3A	LittelFuse	1	pcs	F1	Glass fuse
17	Terminal (Header + Plug)	KF-3.81-2-R	Any	1	pcs	J1	Full
18	USB-C receptacle	USB type C Receptacle	Any	1	pcs	J2	
19	Dsub connector	DB9-Female	Any	1	pcs	J3	
20	Terminal (Header + Plug)	KF-3.81-3-R	Any	1	pcs	J4	Full

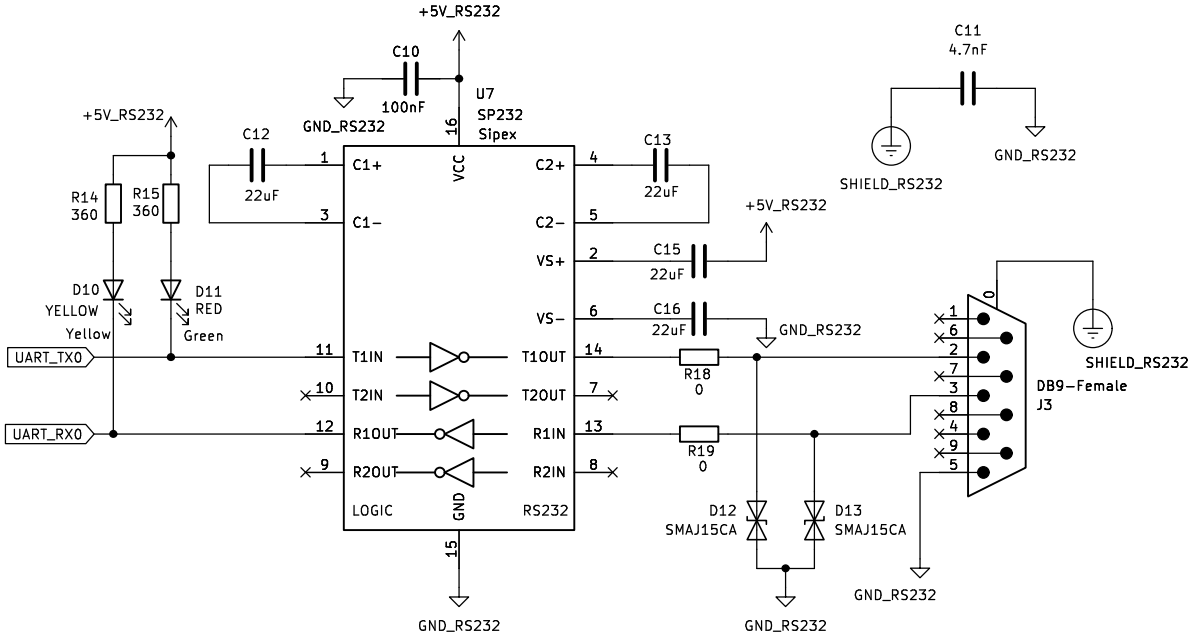
No.	Part Name	Part Code	Maker	Q'ty	Unit	Symbol	Remark
21	Ethernet connector	HR911105A	Hanrun	1	pcs	J5	
22	Terminal (Header + Plug)	KF-3.81-13-R	Any	1	pcs	J6	
23	Terminal (Header + Plug)	KF-3.81-10-R	Any	1	pcs	J7	
24	Inductor	33uH	Any	1	pcs	L1	
25	Inductor	10uH	Any	1	pcs	L2	
26	P-Mosfet	AO3407A	Alpha & Omega Semiconductor	1	pcs	Q1	
27	Transistor	BCX56	NXP	8	pcs	Q2,Q3,Q4,Q5,Q6,Q7,Q8,Q9	
28	Resistor	2k2	Any	11	pcs	R1,R52,R53,R56,R57,R60,R61, R64,R65,RN1,RN2	
29	Resistor	360	Any	11	pcs	R2,R3,R7,R14,R15,R21,R22,R23, R31,RN6,RN7	
30	Resistor	5k1	Any	12	pcs	R4,R5,R6,R12,R16,R20,R26,R27, R28,R33,R36,R37	
31	Resistor	200k	Any	1	pcs	R10	
32	Resistor	68k	Any	1	pcs	R11	
33	Resistor	1k6	Any	1	pcs	R13	
34	Resistor	220	Any	21	pcs	R17,R29,R30,R34,R35,R38,R39, R40,R41,R42,R43,R44,R45,R46, R47,R48,R49,R50,R51,RN5,RN8	
35	Resistor	0	Any	5	pcs	R18,R19,R24,R25,R32	
36	Resistor	510	Any	10	pcs	R54,R55,R58,R59,R62,R63,R66, R67,RN3,RN4	
37	SMD button	SMD 3x6	Any	2	pcs	SW1,SW2	Red
38	Power regulator	LM2596S-5	Texas Instruments	1	pcs	U1	
39	USB TVS diode	USBLC6-2P6	ST Microelectronics	1	pcs	U2	
40	Power regulator	MP1584	Monolithic	1	pcs	U3	
41	Power control	LTC4412	Analog Devices	1	pcs	U4	
42	Isolated power	B0505-1W	Mornsun	2	pcs	U5,U9	

No.	Part Name	Part Code	Maker	Q'ty	Unit	Symbol	Remark
43	High speed optocoupler	HCPL-0600	Fairchild Semiconductor	5	pcs	U6,U8,U11,U12,U13	
44	RS232 Transceiver	SP232	Sipex	1	pcs	U7	
45	RS485 Transceiver	SP485	Sipex	1	pcs	U10	
46	Ethernet converter	CH9121T	WCH	1	pcs	U14	TSSOP20
47	Controller	ESP32-S3-WROOM-1	Espressif Systems	1	pcs	U15	
48	Optocoupler	TLP290-4	Toshiba	6	pcs	U16,U17,U18,U19,U20,U21	
49	Crystal	25MHz	Any	1	pcs	Y1	
50	Plastic box		Maker design	1	pcs		Maker design

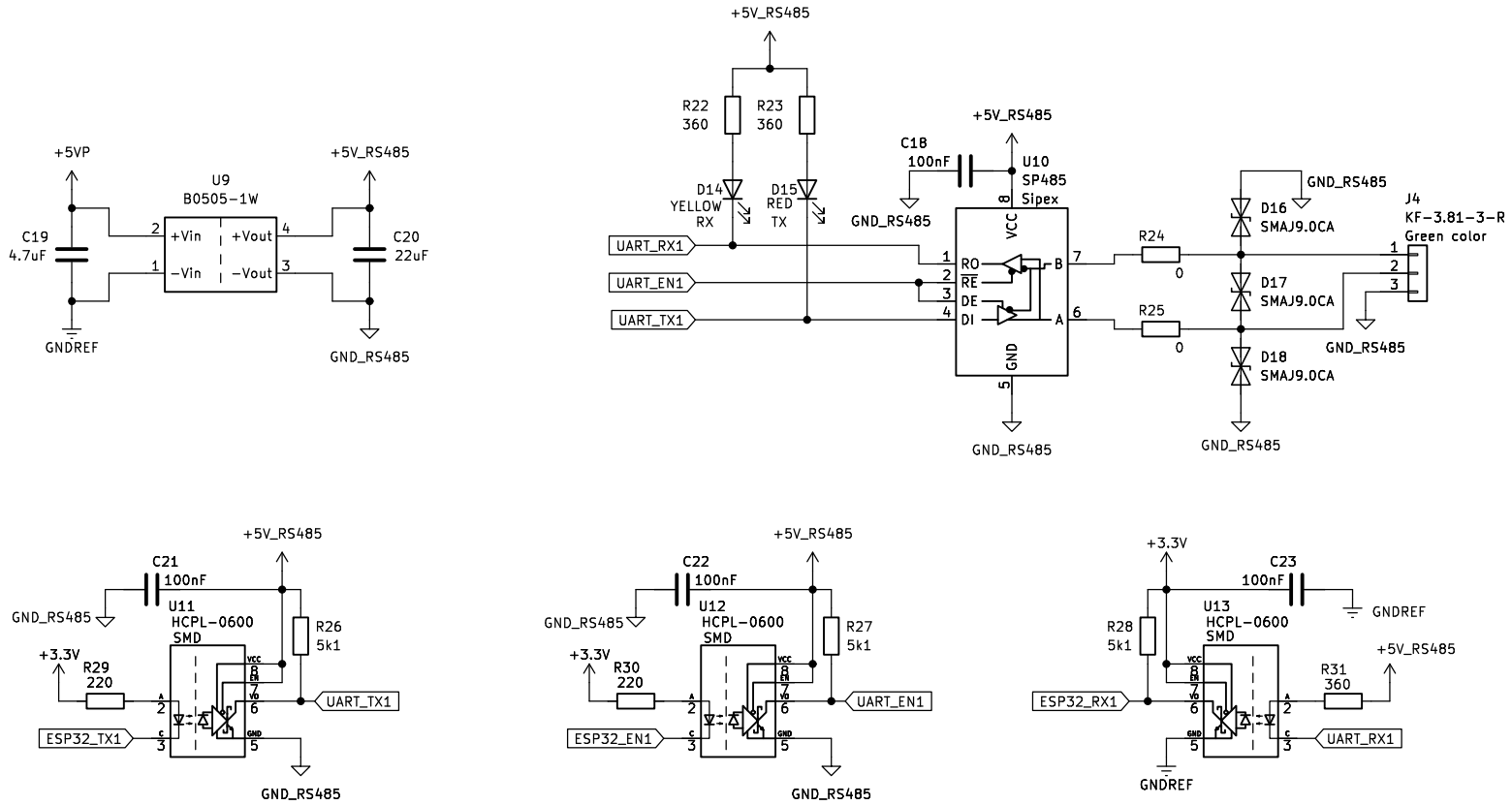


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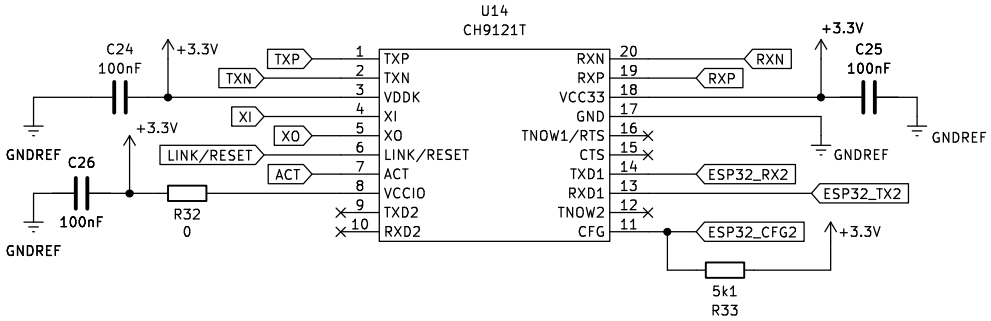


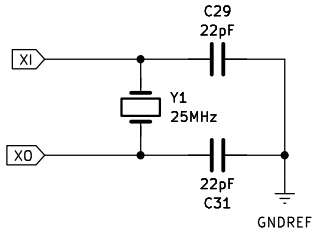


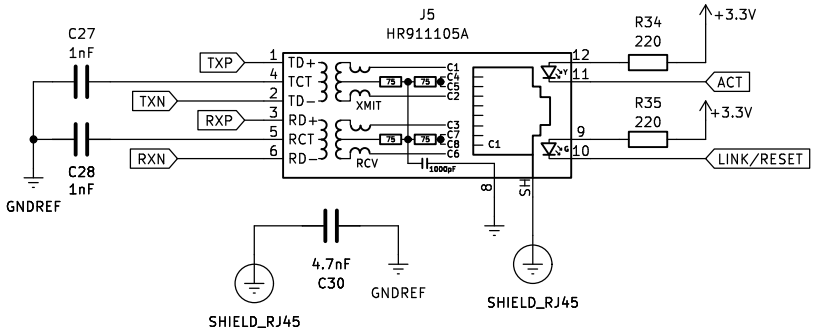
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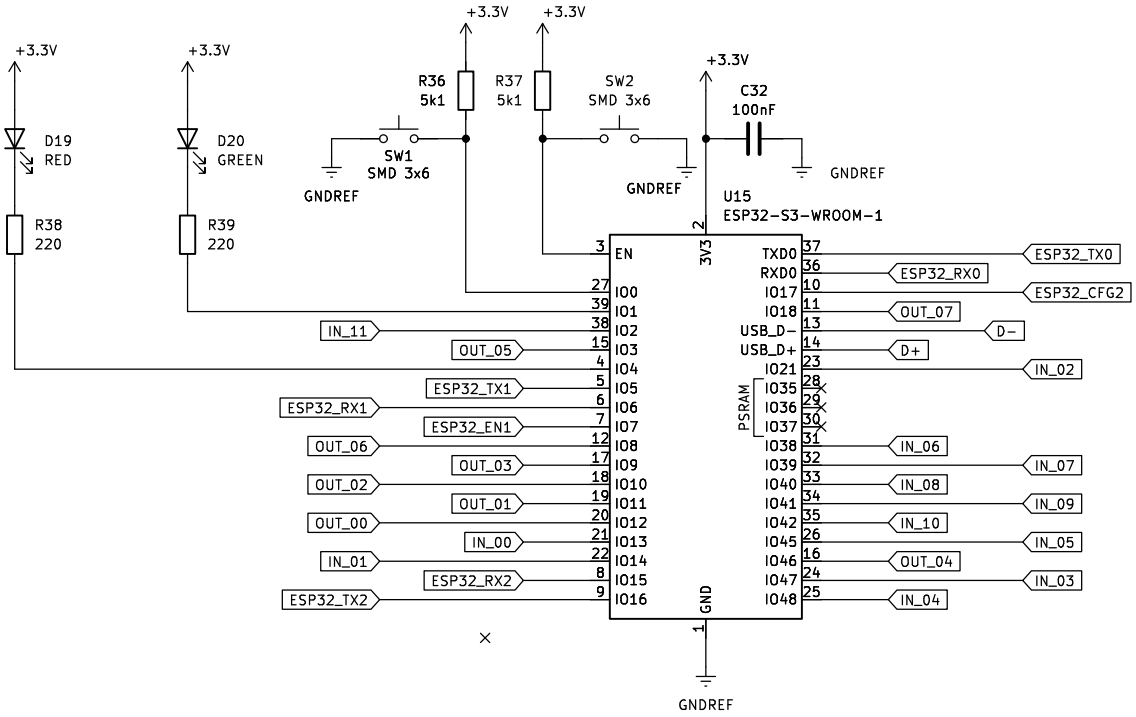
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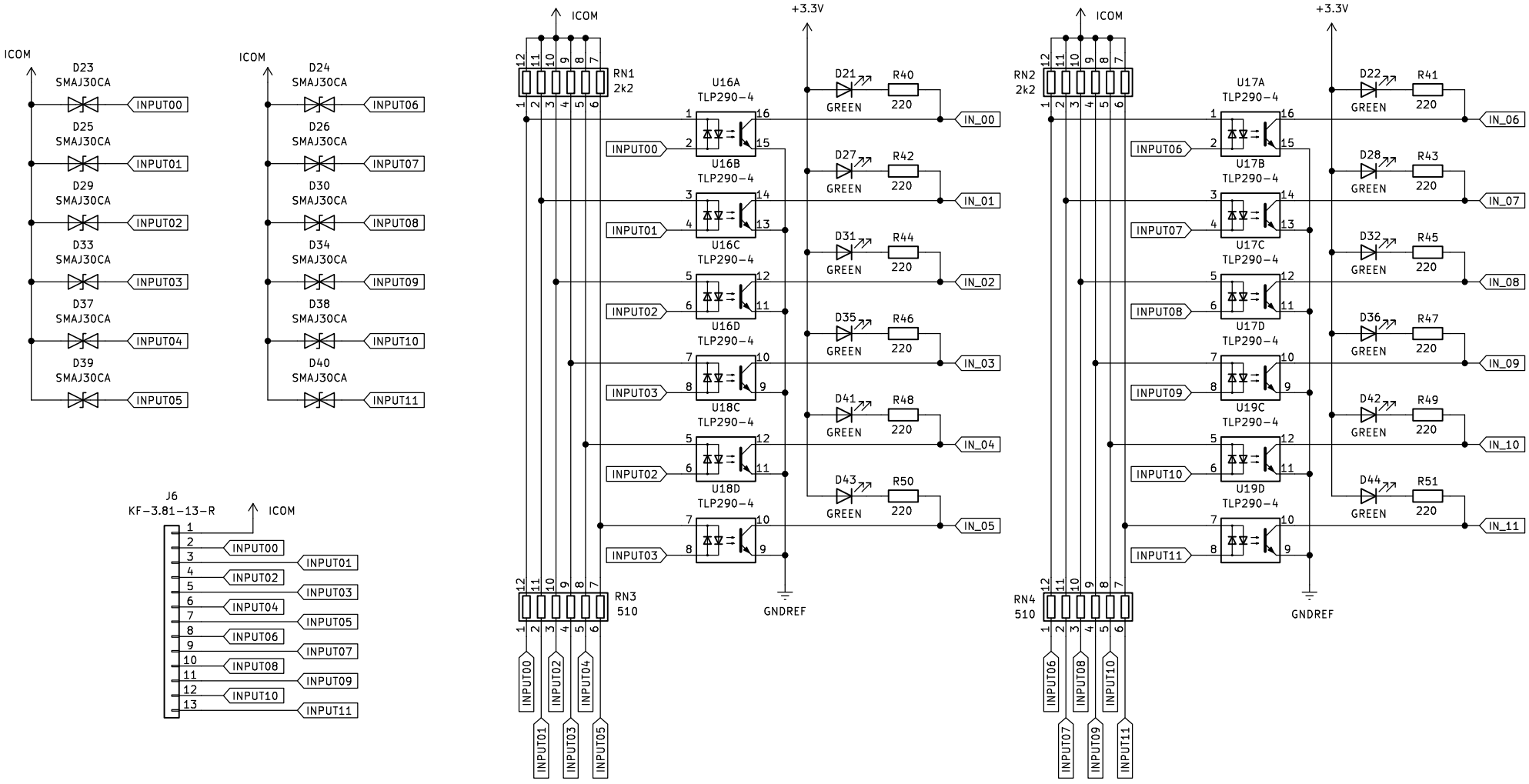




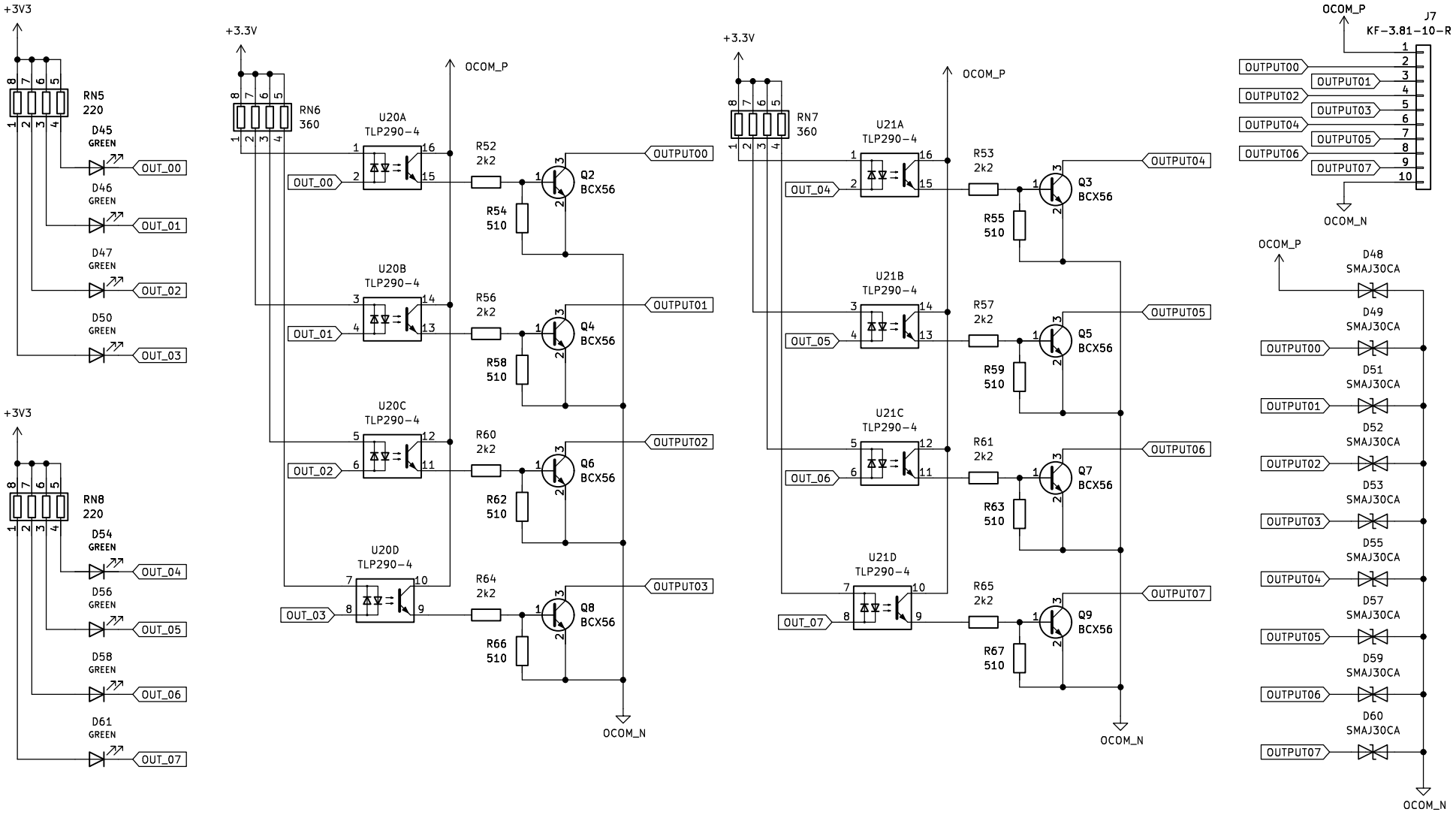
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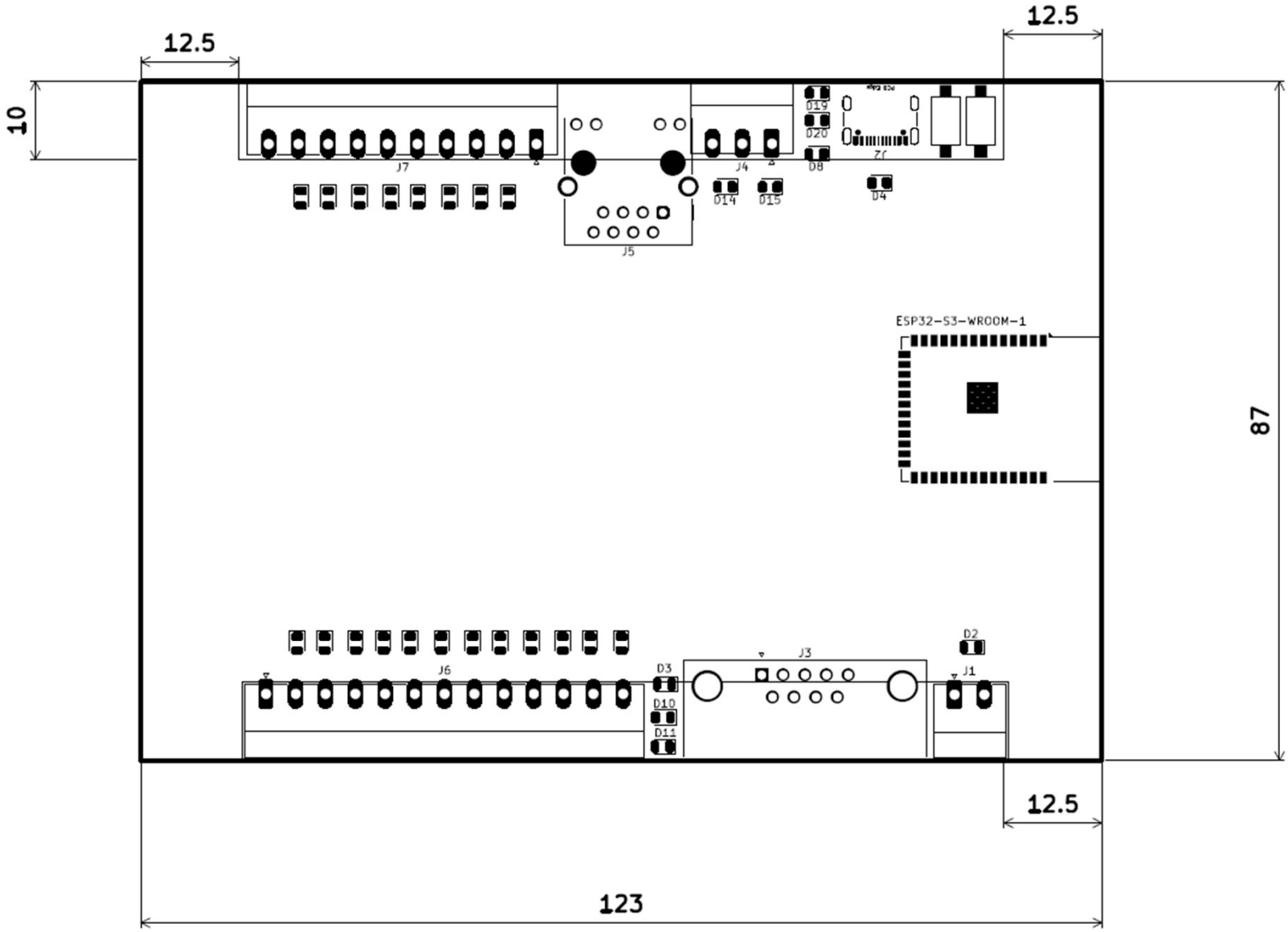
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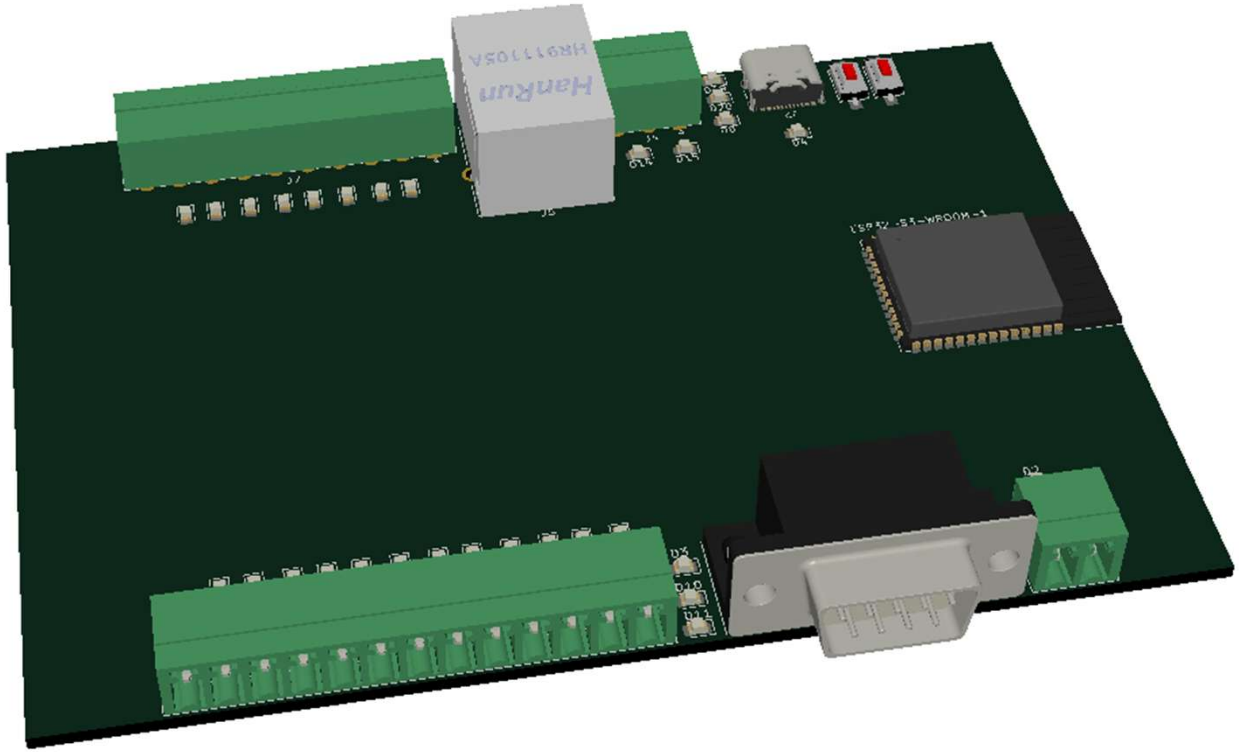
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Note

- Unit: mm
- Maker design and making plastic box

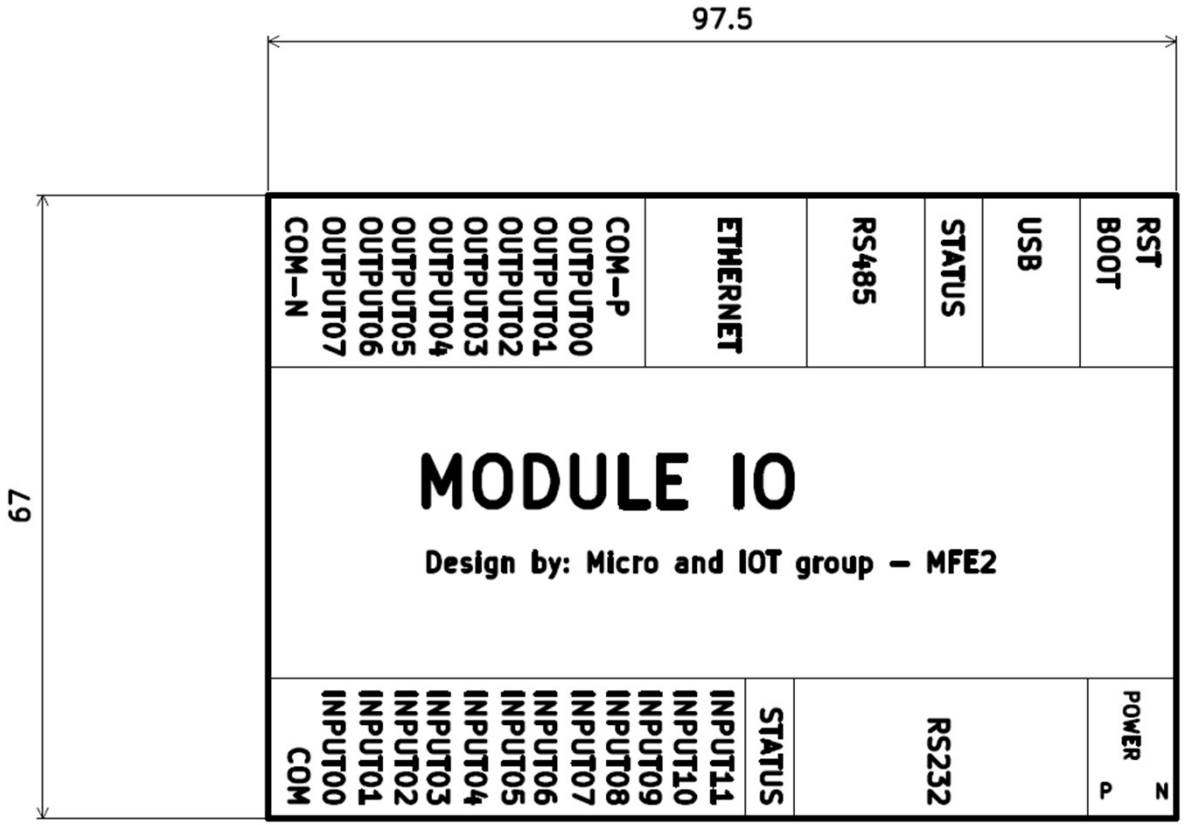
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Note

- Unit: mm
- Maker design and making plastic box

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Note

- Unit: mm
- Maker design and making plastic box

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			N.C.BINH Mar. 2026	MODULE IO QI9-2604-A01 Label	A	01	14